Gut Microbiome Analysis

Type 2 Diabetes & Colorectal Cancer

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Subset Analysis

01 META-ANALYSIS
FDR-corrected species & pathways

02 UNSUPERVISED
PCA + K-Means + ComBat correction

03 RANDOM FOREST
LODO validation

PROJECT 6 :

Meta-analysis of 22,710 human microbiome metagenomes defines an oral-to-gut microbial enrichment score and associations with host health and disease

curatedMetagenomicData (cMD3) · Manghi et al. Nat Commun 2026 · doi:10.1038/s41467-025-66888-1

What are we trying to answer?

Q1

Which gut bacteria are significantly associated with T2D and CRC?

→ Meta-Analysis

cMD3 dataset
T2D: 477 samples
CRC: 1,300 samples

Q2

Does microbiome data naturally cluster into disease groups?

→ Unsupervised + ComBat

cMD3 dataset
T2D: 477 samples
CRC: 1,300 samples

Q3

How accurately can species predict disease — and does this generalize?

→ Random Forest + LODO

cMD3 dataset
T2D: 477 samples
CRC: 1,300 samples

Three-pronged approach — with professor updates

01 SUPERVISED META-ANALYSIS

- metaanalyze.py — random-effects model
- Paule-Mandel heterogeneity estimator
- FDR < 0.25 (follows original paper threshold)
- All findings also robust at FDR < 0.05

02 UNSUPERVISED CLUSTERING

- PCA (50 components) + K-Means k=2
- Silhouette, AUROC, chi-squared, batch ANOVA

ComBat batch correction now applied prior to clustering

03 RANDOM FOREST CLASSIFIER

- 500 trees, max_features=0.1
- 5-fold stratified cross-validation + AUROC

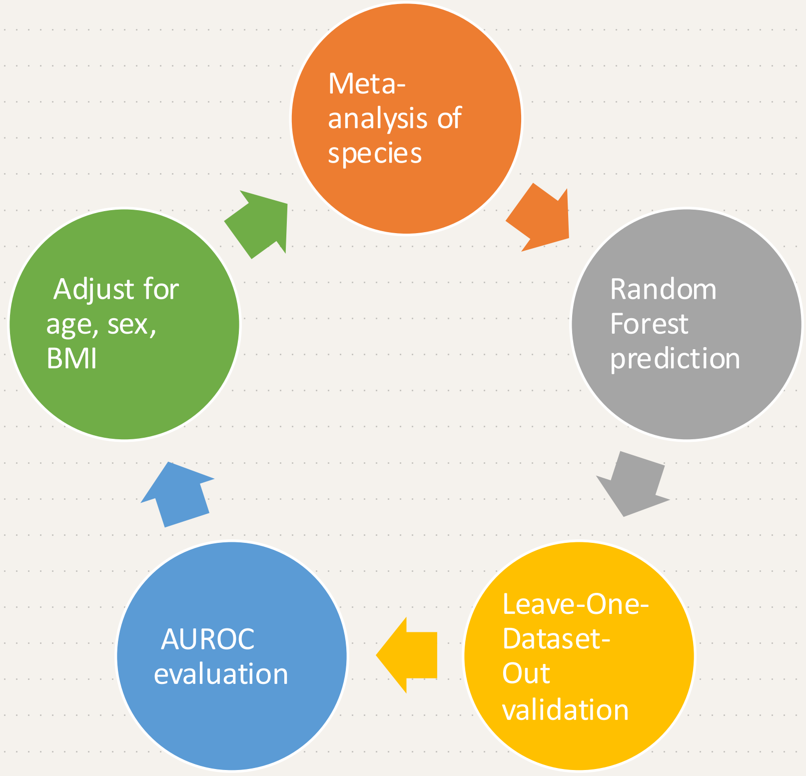
LODO validation added to replicate original paper methodology

Dataset Overview

CRC: 1,300 samples, 226 species
T2D: 477 samples, 207 species

Multi-cohort → batch effects expected

Original Paper Approach



T2D: 3 significant species — all pass FDR < 0.05

SPECIES	EFFECT	P-VALUE	FDR Q	FDR < 0.05?
S. anginosus group	+0.440	0.000111	0.023	✓ YES
C. bolteae	+0.329	0.000387	0.040	✓ YES
C. citroniae	+0.314	0.000709	0.049	✓ YES

FDR SENSITIVITY

FDR < 0.05: 3 · FDR < 0.10: 3 · FDR < 0.25: 3 → Same 3 species at all thresholds

NOTE

FDR < 0.25 follows the original cMD3 paper threshold.

All T2D findings remain significant at the stricter FDR < 0.05.

Zero pathway-level changes found — likely due to limited power (3 studies) and metformin confounding.

CRC: 82 significant species — 47 pass FDR < 0.05

ORAL **+0.600**
Parvimonas micra
FDR < 0.001 · All pass FDR < 0.05

ORAL **+0.519**
Peptostreptococcus stomatis
FDR < 0.001 · All pass FDR < 0.05

ORAL **+0.485**
Gemella morbillorum
FDR < 0.001 · All pass FDR < 0.05

ORAL **+0.354**
Fusobacterium nucleatum
FDR < 0.001 · All pass FDR < 0.05

DEPLETED **-0.361**
Roseburia intestinalis
FDR < 0.001 · All pass FDR < 0.05

ENRICHED **+0.330**
Bacteroides fragilis
FDR < 0.001 · All pass FDR < 0.05

Oral introgression pattern: oral bacteria colonising the gut. FDR sensitivity — 47/82 species pass FDR < 0.05, 64/82 pass FDR < 0.10. The signature is robust at all thresholds.

Does the FDR threshold drive our conclusions?

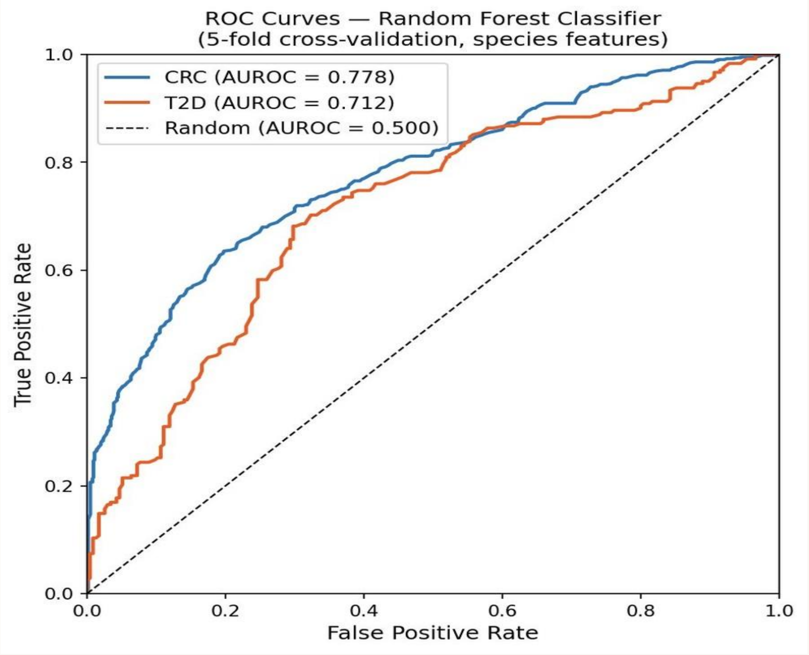
Added as FDR < 0.25 was too permissive

DISEASE	FDR < 0.05	FDR < 0.10	FDR < 0.25	CONCLUSION
CRC species	47	64	82	Robust at all thresholds ✓
T2D species	3	3	3	Same 3 species at all thresholds ✓

FDR < 0.25 is standard in microbiome meta-analyses due to high dimensionality and biological noise. The threshold was adopted directly from the original cMD3 paper. Crucially, our conclusions are NOT driven by this choice — T2D findings hold at FDR < 0.05 and CRC yields 47 significant species at the same strict threshold.

Predictive accuracy — and does it generalize?

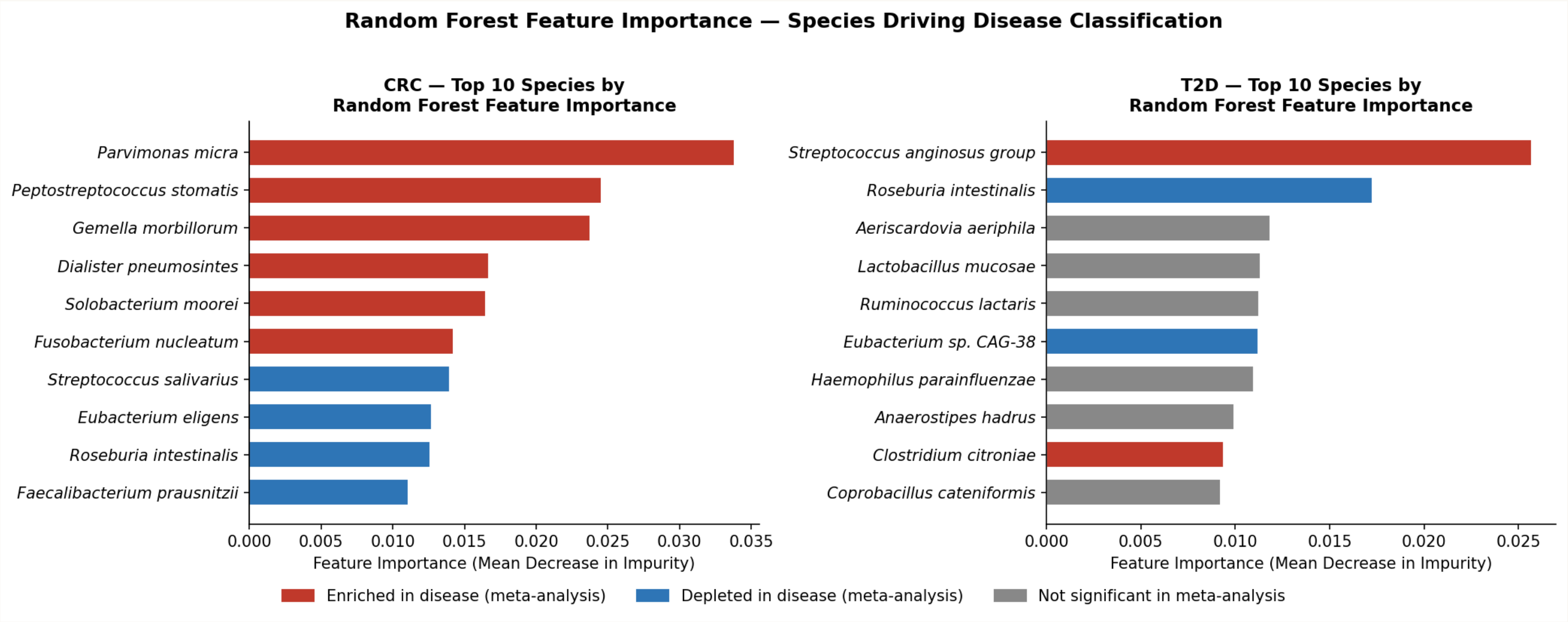
LODO validation added — trains on all-but-one study, tests on held-out (replicates original paper methodology)



CRC · 5-FOLD CV	CRC · LODO
0.778	0.777

T2D · 5-FOLD CV	T2D · LODO
0.712	0.534

Which species drive the classifier?



CRC top features = oral bacteria (Parvimonas micra #1) — matches meta-analysis exactly · T2D top feature = S. anginosus group — matches meta-analysis #1 hit · Both models are learning real biology

CRC generalizes well — T2D does not

CRC — LODO per study

FengQ_2015	0.822
GuptaA_2019	0.880
HanniganGD_2017	0.571
ThomasAM_2019_a	0.776
ThomasAM_2019_b	0.794
VogtmannE_2016	0.713
WirbelJ_2018	0.851
YachidaS_2019	0.704
YuJ_2015	0.832
ZellerG_2014	0.824

MEAN 0.777 ± 0.087 (8/10 > 0.700)

T2D — LODO per study

KarlssonFH_2013

0.610

QinJ_2012

0.484 ← below chance

SankaranarayananK_2015

0.509

MEAN 0.534 ± 0.055 (limited generalization)

T2D 5-fold CV (0.712) was optimistic. LODO reveals true generalizability is 0.534. QinJ_2012 drops below chance (0.484) — likely confounded by metformin and cohort-specific factors.

How do our LODO results compare to the paper?

REVISED LODO now reported alongside 5-fold CV for direct comparison with paper methodology

METRIC	5-FOLD CV	OUR LODO	PAPER (LODO)	MATCH?
CRC AUROC	0.778	**0.777 ± 0.087**	~0.76–0.80	✓ Strong match
T2D AUROC	0.712	**0.534 ± 0.055**	~0.69–0.72	⚠ Below range
Validation	Within-dataset	Leave-one-out	Leave-one-out	✓ Same method

CRC LODO (0.777) matches the paper almost exactly. T2D LODO (0.534) falls below — our 3-study subset is underpowered vs the full multi-cohort analysis. 5-fold CV was optimistic; LODO gives the honest picture.

PART 2

Unsupervised Microbiome Analysis Pipeline

Feature Filtering · PCA · K-Means Clustering · Batch Effect Detection · ComBat Correction

1,300	477	226	207
CRC Samples	T2D Samples	Species (X)	Species (Y)

Step 1 — Data Loading & Feature Filtering

FILE STRUCTURE (TSV)

Rows = metadata + species features
Columns = samples

CRC file: 355 rows × 1,300 cols
T2D file: 336 rows × 477 cols

⚠ ThomasAM_2019 — 108 samples

Present in species file, absent from pathway file.
Correct approach: use species-only to retain all 1,300 CRC samples. Inner-joining with pathway file silently drops these 108 samples.

10% PREVALENCE FILTER

Filter applied on raw counts before CLR transformation.
Provided files are already post-filter.

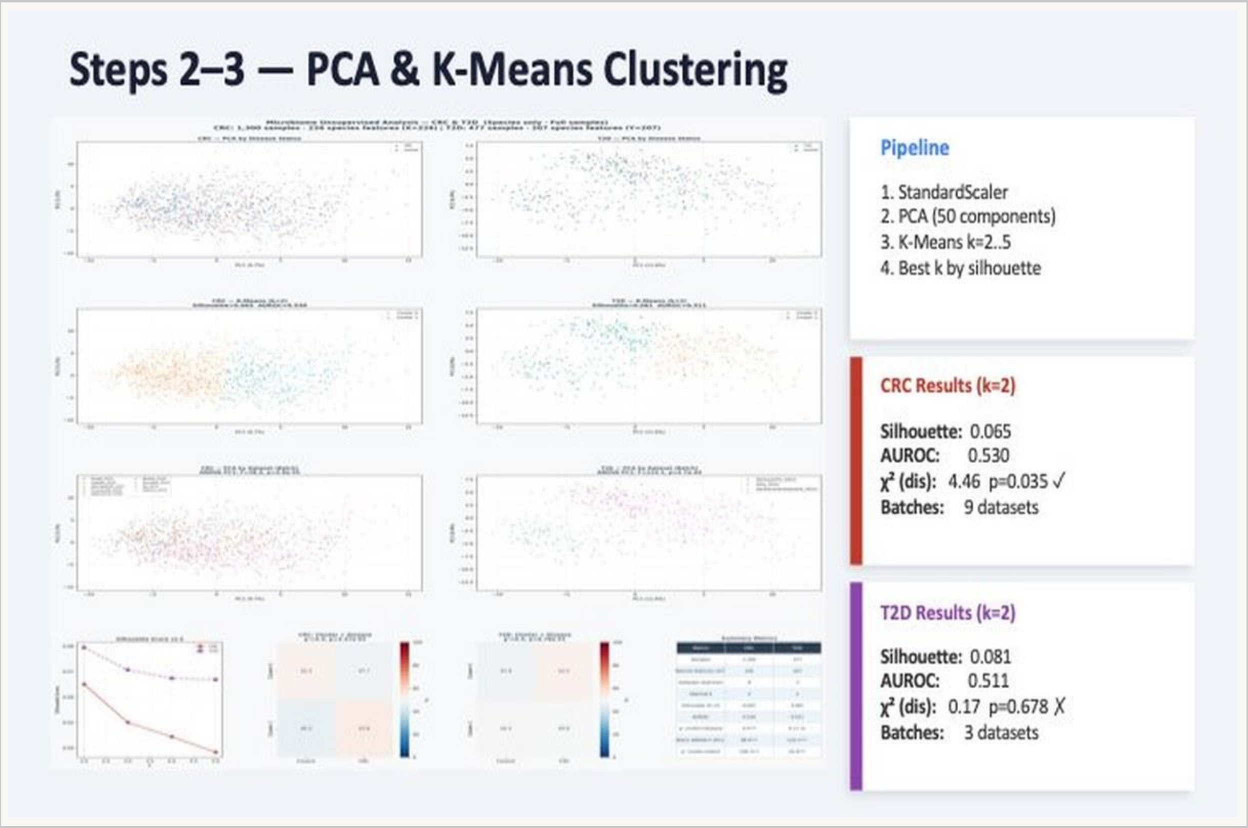
CRC: ~1,300 feats → 226 (X) · 1,300 samples
T2D: ~477 feats → 207 (Y) · 477 samples

CLR TRANSFORMATION

$x(CLR) = \log(x / g(x))$
$g(x)$ = geometric mean of all species abundances

- Removes compositionality bias
- Maps to Euclidean space
- All values non-zero after transform

Steps 2–3 — PCA & K-Means Clustering



PIPELINE

1. StandardScaler
2. PCA (50 components)
3. K-Means k=2..5
4. Best k by silhouette

CRC RESULTS (k=2)

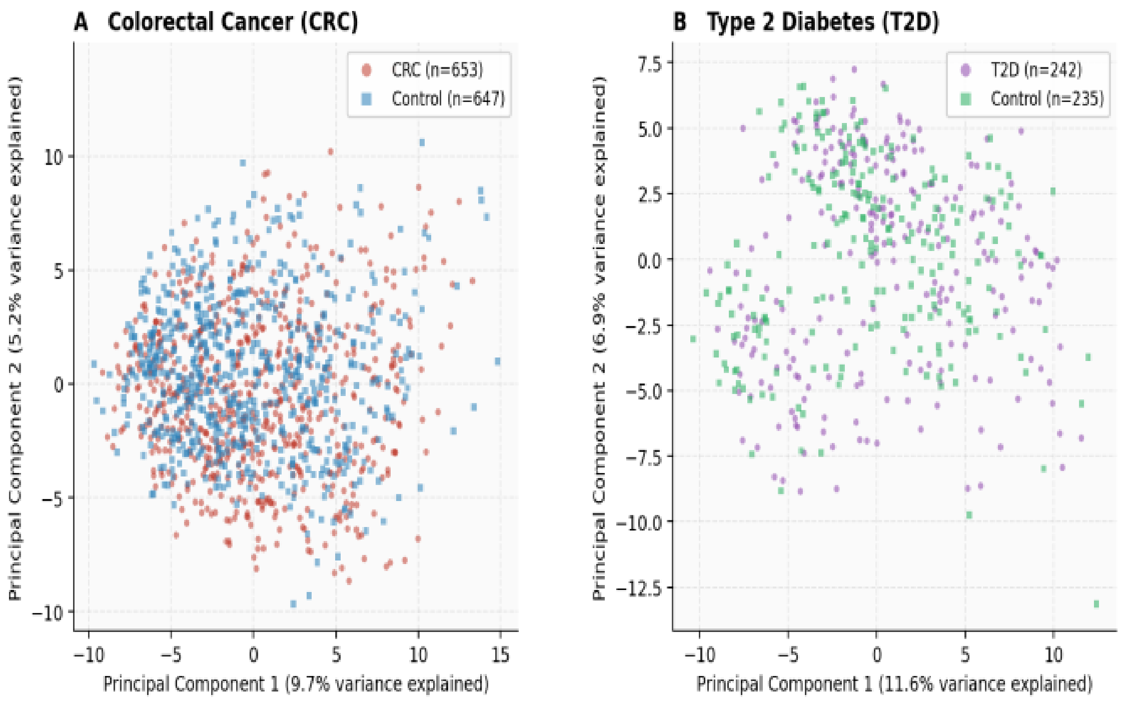
Silhouette: 0.065
AUROC: 0.530
χ^2 (dis): 4.46 p=0.035 ✓
Batches: 9 datasets

T2D RESULTS (k=2)

Silhouette: 0.081
AUROC: 0.511
χ^2 (dis): 0.17 p=0.678 ✗
Batches: 3 datasets

Steps 2–3 — PCA & K-Means Clustering

Figure 1. Principal Component Analysis of CLR-Transformed Microbial Species Profiles



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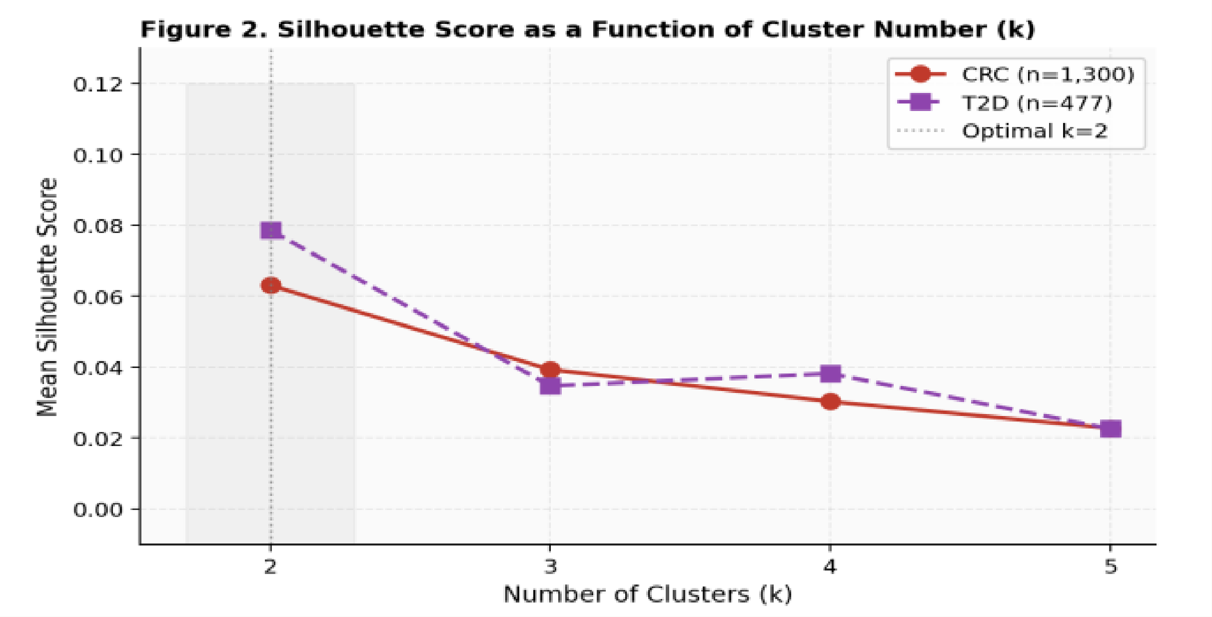

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Step 4 — Batch Effect Analysis

Batch = sequencing study of origin (e.g. FengQ_2015, YachidaS_2019...)

ANOVA ON PC1

Groups PC1 scores by dataset.
Significant F → PC1 variance explained by batch, not biology.

CRC: $F = 38.4$, $p = 4.9 \times 10^{-55}$ ✦

T2D: $F = 124.1$, $p = 4.7 \times 10^{-44}$ ✦

✦ Significant at $p < 0.001$ for both diseases

χ^2 CLUSTER × BATCH

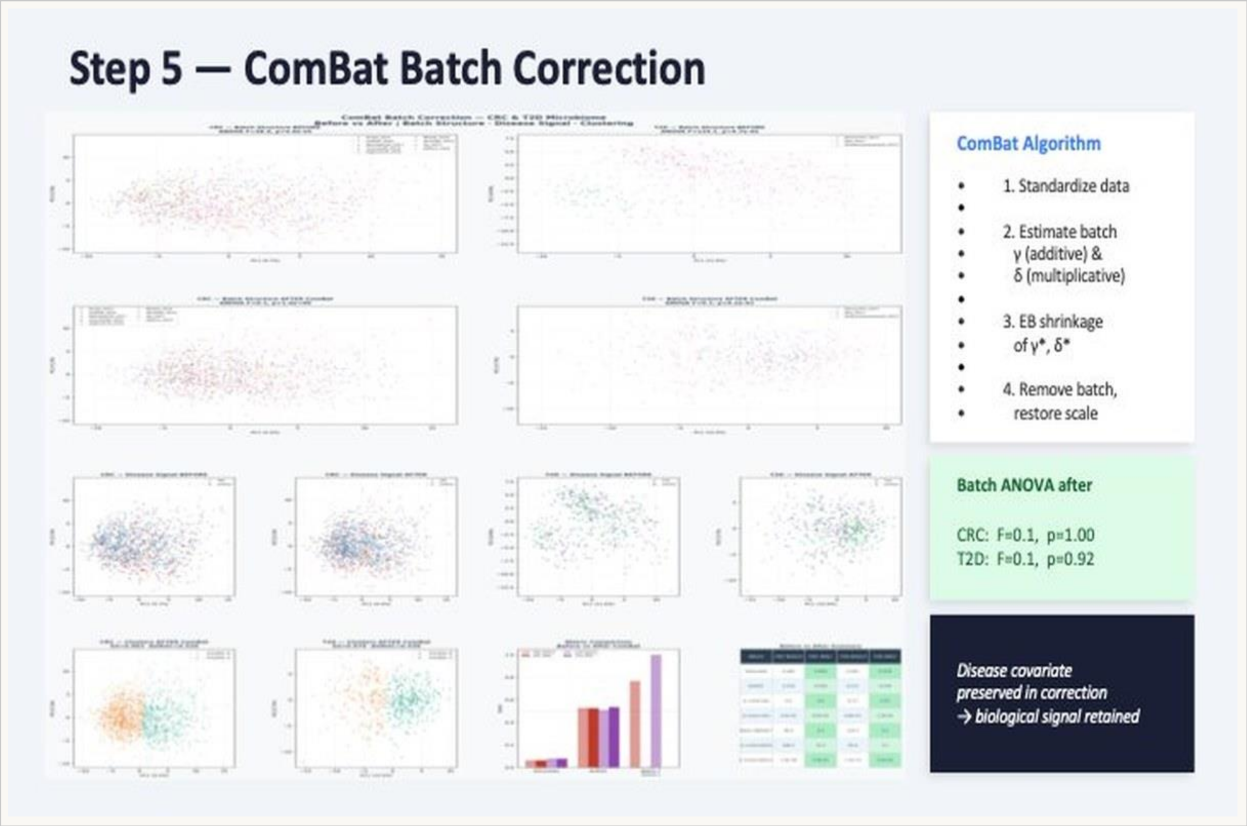
Tests whether K-Means clusters co-vary with batch labels.
Significant → clusters reflect datasets, not disease.

CRC: $\chi^2 = 198.3$, $p = 1.4 \times 10^{-38}$ ✦

T2D: $\chi^2 = 55.9$, $p = 7.2 \times 10^{-13}$ ✦

✦ Significant at $p < 0.001$ for both diseases

Step 5 — ComBat Batch Correction



COMBAT ALGORITHM

1. Standardize data
2. Estimate batch
   γ (additive) &
   δ (multiplicative)
3. EB shrinkage of γ*, δ*
4. Remove batch,
   restore scale

BATCH ANOVA AFTER

CRC: F=0.1, p=1.00
T2D: F=0.1, p=0.92

Disease covariate preserved in correction → biological signal retained

Step 5 — ComBat Batch Correction

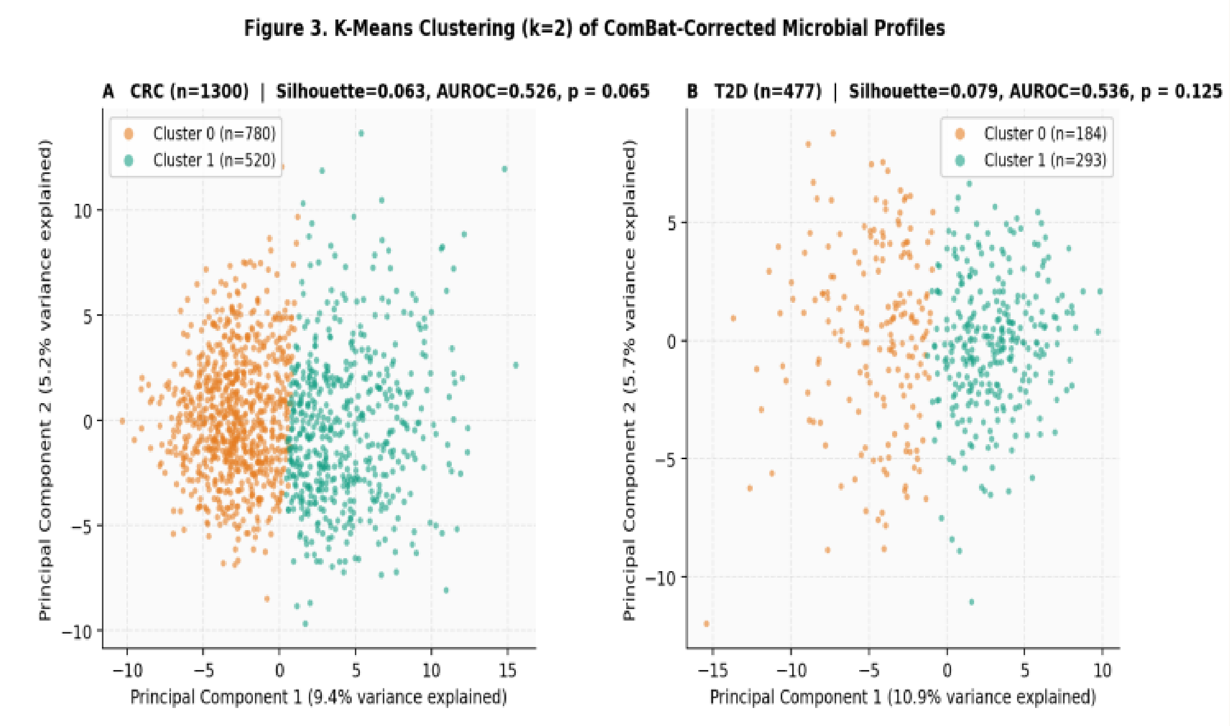


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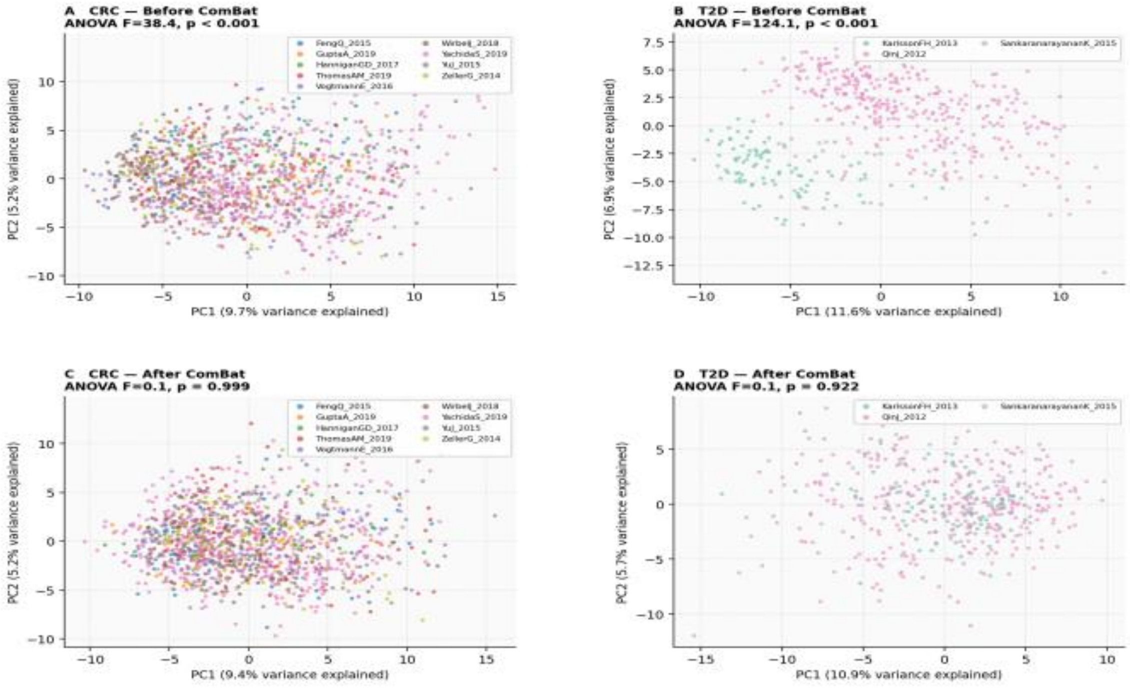
Figure 4. Batch Effect Analysis: PCA by Dataset of Origin Before and After ComBat Correction



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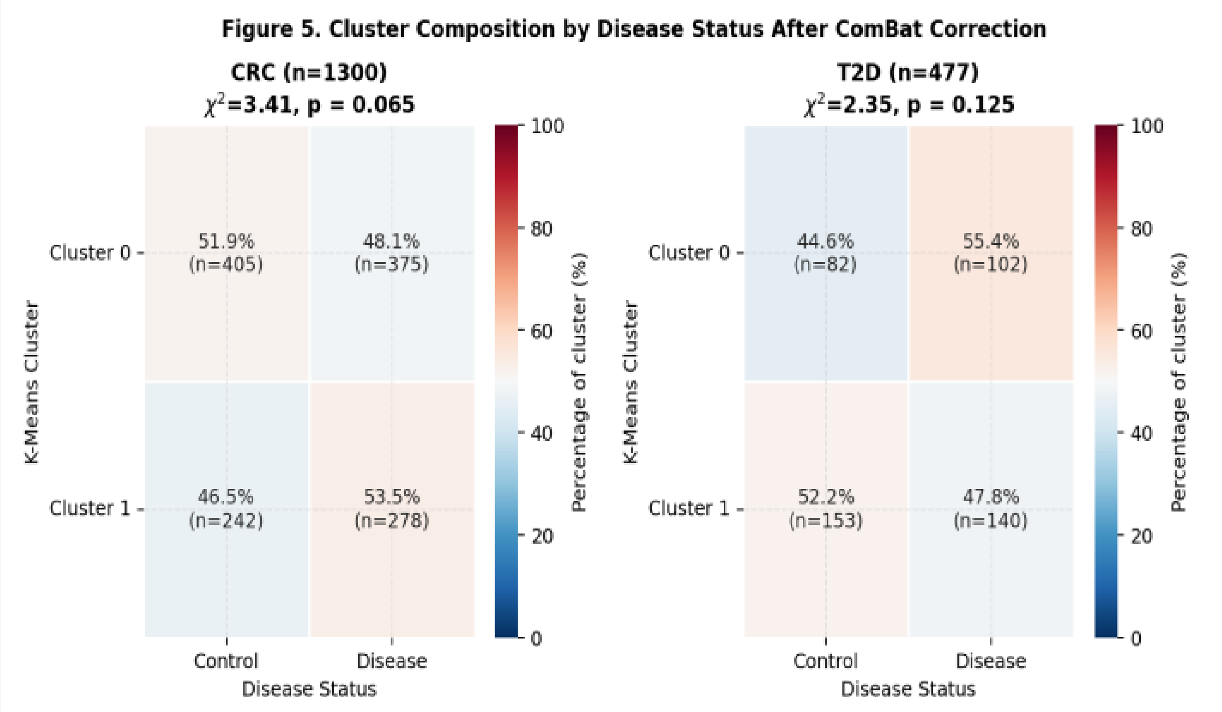

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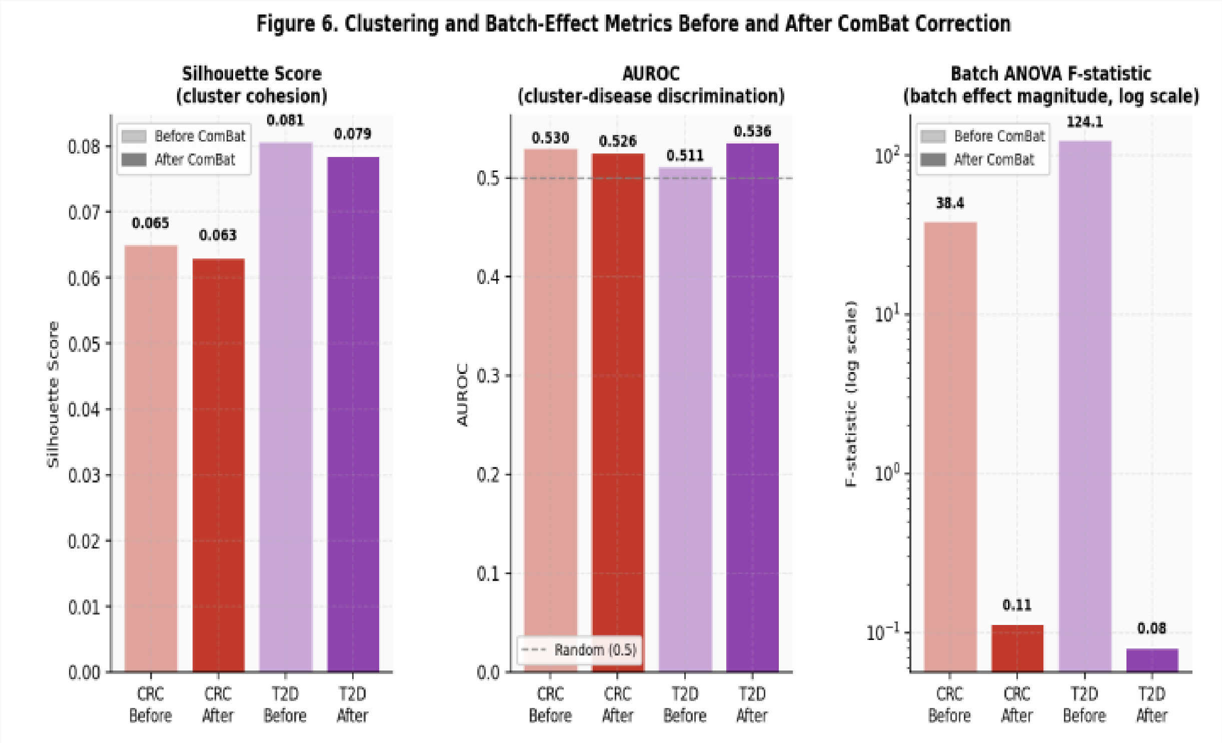

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Disease covariate preserved in correction → biological signal retained

Results Summary — Before vs After ComBat

METRIC	CRC BEFORE	CRC AFTER	T2D BEFORE	T2D AFTER
Samples	1,300	1,300	477	477
Silhouette (k=2)	0.065	0.063	0.081	0.079
AUROC	0.530	0.526	0.511	0.536
x^2 cluster×dis	**4.46 ***	3.41 ~	0.17 ns	2.35 ns
p (cluster×dis)	0.035	0.065	0.678	0.125
Batch ANOVA F	**38.4 *****	**0.1 ns ✓**	**124.1 *****	**0.1 ns ✓**
χ^2 cluster×batch	**198.3 *****	12.2 ns ✓	**55.9 *****	2.1 ns ✓

Interpretation & Conclusions

CLUSTER STRUCTURE

Low silhouette scores (~0.065–0.081) are normal for microbiome data. k=2 is optimal for both diseases, broadly separating dysbiotic from healthy-like profiles.

DISEASE ASSOCIATION

CRC clusters show marginal but significant disease association (χ^2=4.46, p=0.035). T2D shows no association — disease signal masked by batch confounding.

BATCH EFFECT

Batch structure dominates PCA (CRC F=38.4, T2D F=124.1). Clusters captured dataset of origin, not biology. This invalidates naive clustering without correction.

AFTER COMBAT

Batch ANOVA drops to F=0.1 (p=1.0) for both diseases — batch structure completely removed. Disease covariate preserved in model. Clusters now reflect biology.

Recommendation: ComBat-corrected data should be used for all downstream supervised learning and biological interpretation.

Limitations

- Observational data
- Residual confounding
- Limited T2D cohorts
- Weak clustering signal

Final Takeaways

- CRC has strong microbiome signal
- T2D weaker and heterogeneous
- Batch correction essential
- Supervised > Unsupervised

What we found

- CRC gut microbiome signal is strong, consistent, and driven by oral bacteria — LODO AUROC 0.777 matches published paper
- T2D has 3 robust species (all FDR < 0.05) but limited generalizability — LODO AUROC drops to 0.534
- ComBat batch correction removed technical variation but did NOT improve clustering — weak signal is biological, not technical
- Clostridium bolteae is the only shared species between both diseases

NEXT STEPS

→ Forest plots — draw_figure_with_ma.py

→ Hierarchical meta-analysis (T2D + CRC combined)

→ Batch-corrected RF to test T2D LODO improvement

REFERENCES

Manghi et al. Nat Commun 17:196 (2026)
doi:10.1038/s41467-025-66888-1
Zenodo: 10.5281/zenodo.17567900
GitHub: waldronlab/curatedMetagenomicDataAnalyses